DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT25-425 P-II
COMPUTER NETWORKS
MORNING/EVENING

LAB#7

TASK: STUDY OF ROUTER AND ROUTER INTERACE CONFIGURATION COMMANDS.

Name: _____ Roll No:_____

Score: _____ Signature of the Lab Tutor _____ Date: _____

BASIC NETWORK AND CONFIGURATION COMMANDS

Objectives:

Study of basic network command and Network configuration commands

Apparatus (Software): Command Prompt And Packet Tracer.

Procedure: To do this EXPERIMENT- follows these steps:

In this EXPERIMENT- students have to understand basic networking commands e.g ping,tracert etc.

All commands related to Network configuration which includes how to switch to privilege modeand normal mode and how to configure router interface and how to save this configuration to flash memory or permanent memory.

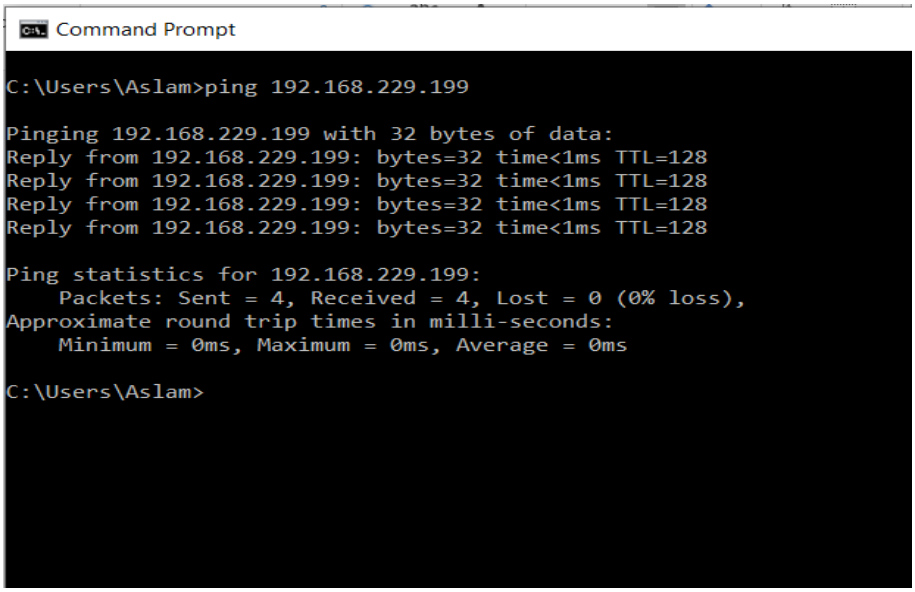
This commands includes

- Configuring the Router commands
- General Commands to configure network
- Privileged Mode commands of a router
- Router Processes & Statistics
- IP Commands
- Other IP Commands e.g. show ip route etc.

Ping:

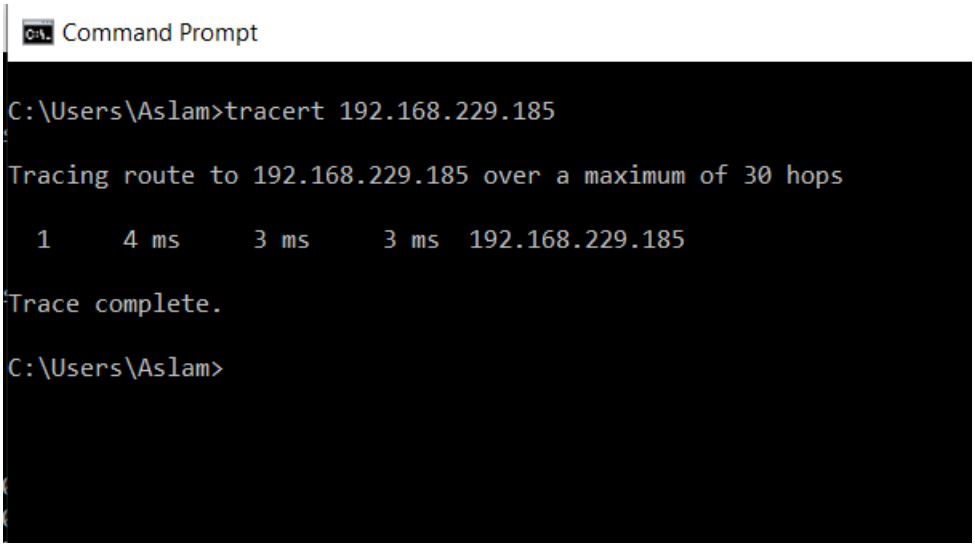
ping sends an ICMP ECHO_REQUEST packet to the specified host. If the host responds, youget an ICMP packet back. Sound strange? Well, you can “ping” an IP address to see if a machineis alive. If

there is no response, you know something is wrong.



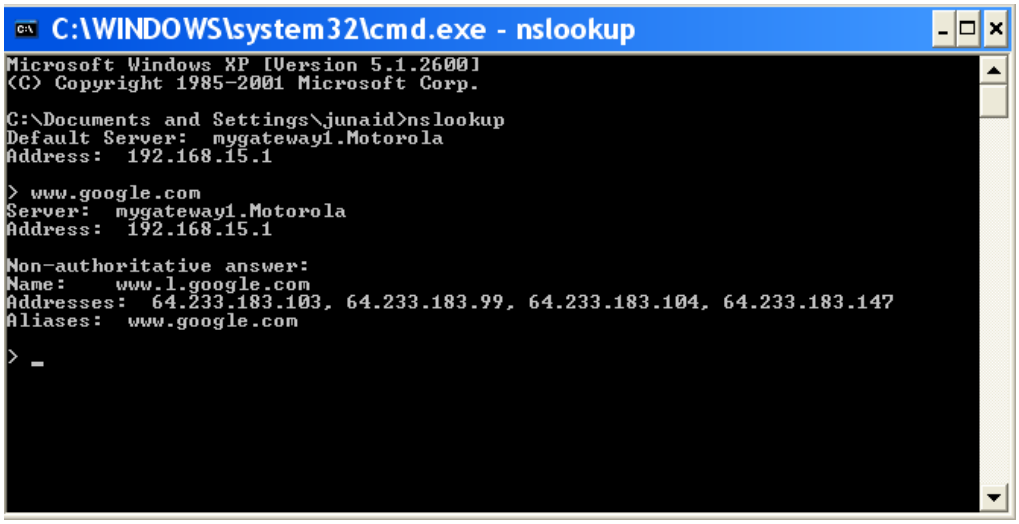
Traceroute:

Tracert is a command which can show you the path a packet of information takes from your computer to one you specify. It will list all the routers it passes through until it reaches its destination, or fails to and is discarded. In addition to this, it will tell you how long each 'hop' from router to router takes.



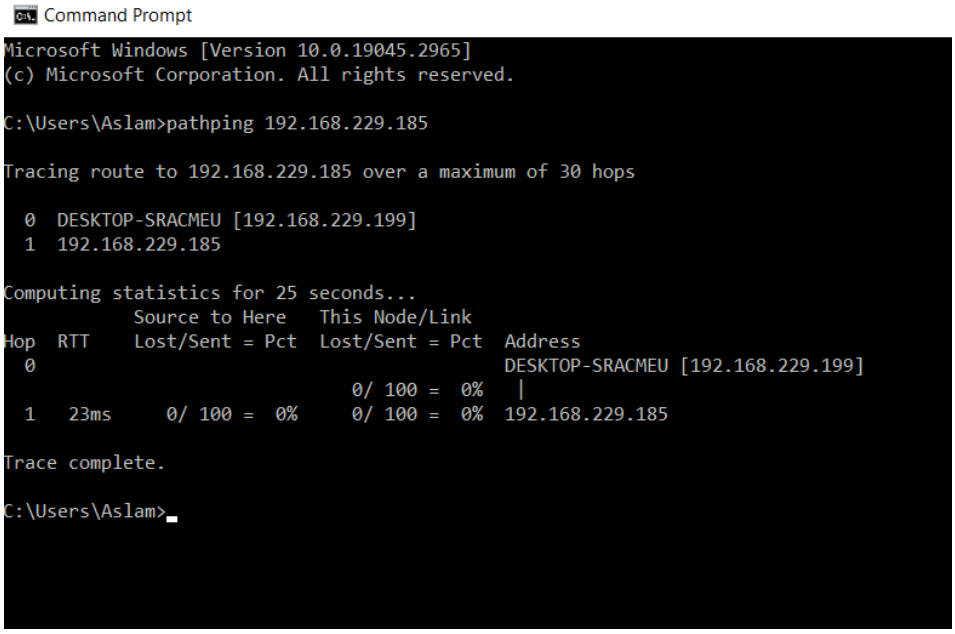
nslookup:

Displays information from Domain Name System (DNS) name servers.
NOTE :If you write the command as above it shows as default your pc's server name firstly.

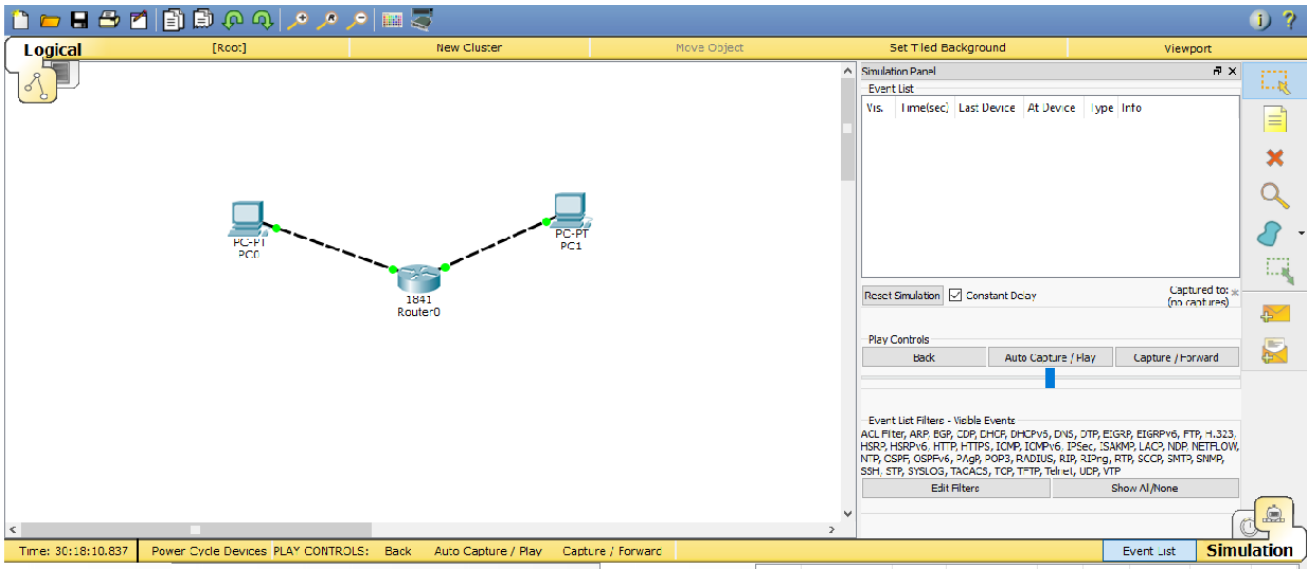


pathping:

A better version of tracert that gives you statics about packet lost and latency.



First of all, we setup a simple network consisting routers and to PCs for this we need configure router with main line interface to make it work.



Configuration: Click on the router0, go to the mainline interface and click CLI tab.
Continue with Configuration dialog? (yes/no) No
Press No because we will manually configure the router with commands.
To enter into the privileged exec mode type:
Router> enable
You will enter into the privileged exec mode when you see router followed by # sign
Router#
To enter in to the global configuration mode type:
Router #config t
To name the router type:
Router(config)#hostname Router0
To set an encrypted password for logging in to the privileged exec mode type:
Router0(config)#enable secret class
(In real world situation class is obviously not an acceptable password, but to perform experiment through packet tracer we will write either the class or cisco)
Now to configure the password on console line type:
Router0(config)#line con 0
Then to set the password type:
Router0(config-line)#password cisco
To enable login to make password protected type:
Router0(config-line)#login
Press exit to return in global configuration mode
Router0(config-line)#exit
Router0(config)#
To configure the password for virtual terminal lines by entering:
Router0(config)#line vty 0 4
Router0(config-line)#
The **VTY lines** are the Virtual Terminal **lines** of the router, used exclusively to control Telnet connections. They are virtual, in the sense that they are a function of software - there is no hardware associated with them.

Router0(config-line)#exit
Router0(config)#

Now, to configure the fast Ethernet interfaces, for this write the interface and then the interface name. In this case we start with FastEthernet0/0

Router0(config)#interface FastEthernet 0/0

To set this interface's IP Address and Subnet Mask Type:

Router0(config-if)#IP Address 192.168.1.1 255.255.255.0

For the optional you may want to add description for the later reference type:

Router0(config-if)#description Router0 FastEthernet0/0

Now, enable the interface type:

Router0(config-if)#no shutdown

Exit back to global exec mode type:

Router0(config-if)#exit

Repeat same process for Fast Ethernet for 0/1 interface type:

Router0(config)#interface FastEthernet0/1

Router0(config-if)#IP Address 192.168.2.1 255.255.255.0

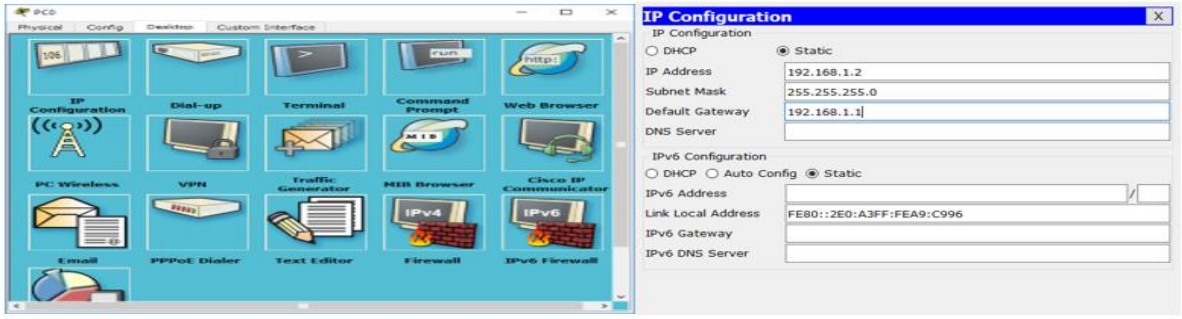
Now, enable the interface type:

Router0(config-if)#no shutdown

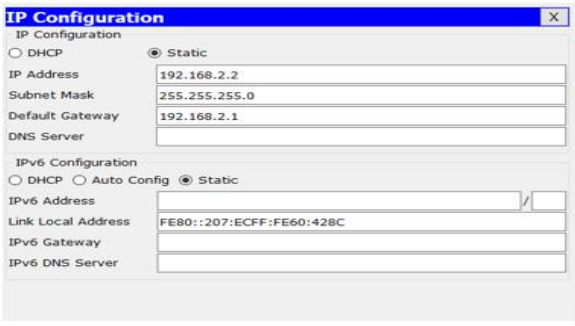
Exit back to interface mode type:

Router0(config-if)#exit

To Configure the PC first click at PC0, go to the desktop tab, press IP Configuration:



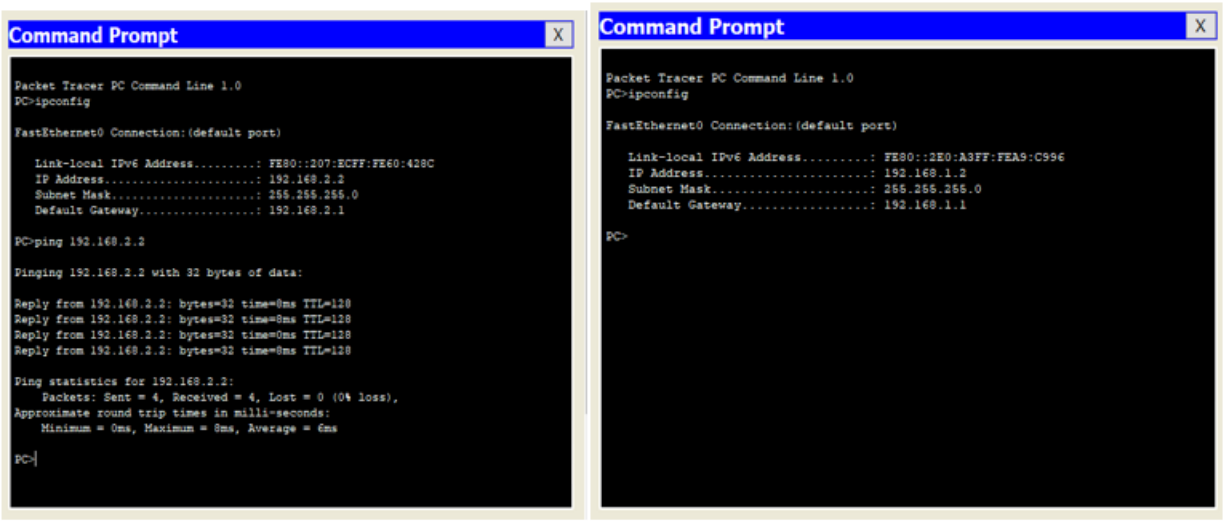
Then click PC1 to set the IPs as fallows



Now check the PC configuration to check connectivity through ping, click PC0 and then click command prompt.



To check the IP address configuration and ping type:



It seems that ping is successful

To lets visually see the ping with simulation mode, click bottom right side corner behind the Real-time tab click to activate simulation mode.

Exit back to global exec mode type:
Router0(config)#exit
Hit on bellow prompt and you will be back at the privileged exec mode
%SYS-5-CONFIG_I: Configured from console by console
Router0#
To display the current configuration type
You will all configuration that you have set
Router0#show running-config
Building configuration...

Current configuration : 650 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router0
!
enable secret 5 1mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip cef
no ipv6 cef
!
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0

duplex auto
speed auto
!
interface FastEthernet0/1
ip address 192.168.2.1 255.255.255.0
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
ip flow-export version 9
!
line con 0
password cisco
login
!
line aux 0
!
line vty 0 4
login
!
end
To save this configuration type
Router0#copy running-config startup-config

To confirm hit bellow prompt

Destination filename [startup-config]?
Building configuration...
[OK]

Review Questions

Question 1: What is the basic command to access the global configuration mode on a Cisco router?

Answer:

Question 2: What is IOS?

Answer:

Question 3: What do you mean by User mode and Privileged mode?

Answer:

Note: Submit the handout before next lab